Autonomous Research System — Stage 1

A controlled, reproducible system with explicit execution boundaries and upgradeable architecture.

Harness-Mediated Autoresearch Runtime · April 2026 · Internal One-Pager

What Was Built

A control-plane governed system for autonomous experimentation, where a planning model supervises bounded local execution under explicit lifecycle and memory constraints.

Target application: optimizing a **ResNet-based binary trigger** in a near-threshold, high-noise regime where performance is sensitive to small hyperparameter changes.

Why this matters

Turns experimentation into a controlled, auditable, repeatable process — required for reproducible and controlled experimentation.

Designed for **local / private execution** (no data leaves the machine), **multi-node scaling**, and **model-agnostic orchestration**.

Stabilizes experimentation under noisy regimes through controlled iteration — each run is bounded, reproducible, and comparable.

Prevents data leakage and protects code / IP via local bounded execution — no external API, no uncontrolled file access.

Architectural direction: a **reusable system abstraction for controlled experimentation**, not a single-task agent.

Campaign Result — 3 Manager Rounds × 5 Worker Experiments

Metric	Value	Notes
Baseline AUC	**0.7799**	Starting point
Best AUC	**0.7876**	R3W4: LR=5e-4, WD=3e-5, dropout=0.02
Improvement	**+0.77%**	Fully autonomous — no manual intervention
Kept / discarded	5 / 10	Every decision logged with rationale
Execution	Local CPU only	Ollama + Qwen 2.5 7B Coder — no external API

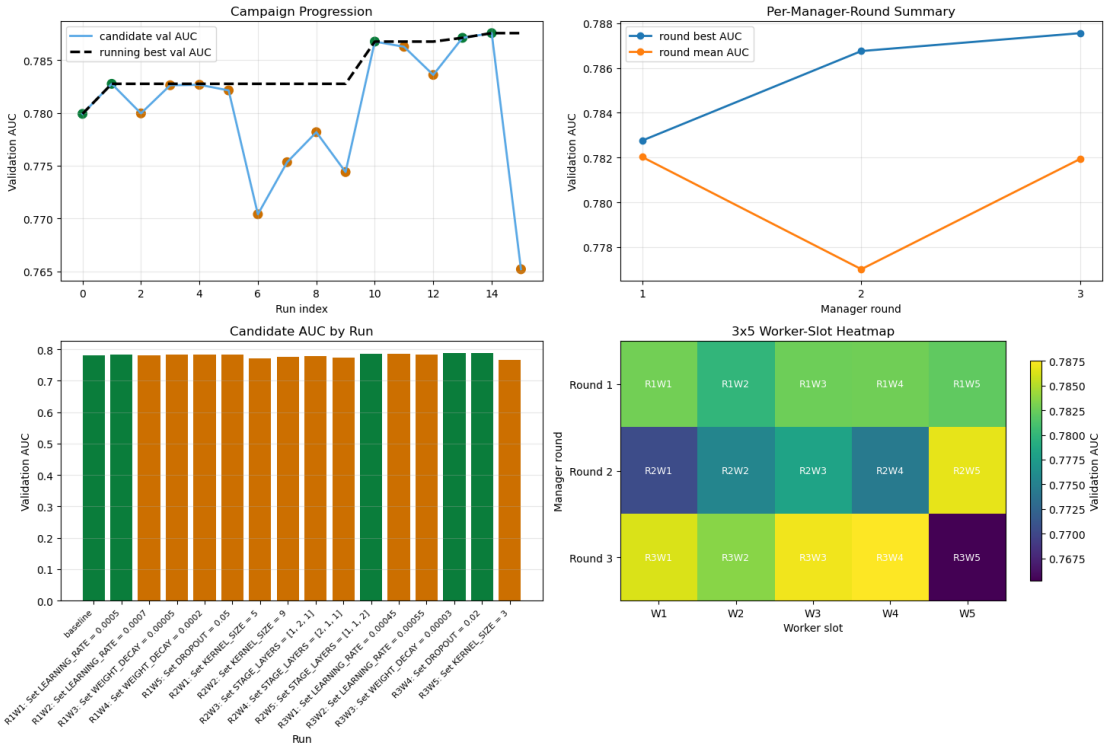

Figure 1. Campaign progression, per-round summary, per-run bar chart, and worker-slot heatmap.

Architecture (3-Layer Harness)

Layer	Role	Key property
Control Plane	Governance / harness runtime	Owns state machine, memory, keep/discard lifecycle — no agent acts unsupervised
Manager	Planning / reasoning	Reads memory, proposes bounded experiment, issues keep/discard with rationale
Worker	Execution (bounded)	Edits only train.py — hard assertion at commit; every perf change attributable to one diff

Next Steps

Stage 2: Automated /loop execution — 5-round × 10-worker overnight campaign.
Stage 3: Multi-node and parallel branch execution using the same control plane API.
Stage 4: Domain integration — detector filter tuning, EMPCA basis, threshold optimization.

The key contribution is demonstrating that a **governed, harness-mediated system can produce reliable and auditable research progress under strict execution constraints** — forming the foundation for a scalable and controlled research workflow.